

Audio course on steroids during pregnancy. Episode five. PPROM, tocolysis, preeclampsia, and the steroid-completion trap.

Companion reading handout for simultaneous listening.

This episode is about time.

Steroids need time to help the fetus. But not every pregnancy can safely buy time. PPROM, infection, bleeding, severe preeclampsia, HELLP, and fetal compromise are all settings where the resident must ask whether waiting helps or harms.

The trap is to treat steroid completion as a goal by itself.

Steroid completion is not the goal. Safe maternal-fetal care is the goal. Steroid completion is valuable only when time is safe.

Start with PPROM.

PPROM creates a classic obstetric tension. The fetus may benefit from more maturity, and antenatal corticosteroids may reduce complications of prematurity. But ruptured membranes also increase concern for ascending infection, cord complications, and the need for delivery if chorioamnionitis appears.

The Israeli PPROM position paper recommends corticosteroids from viable gestational age at least until thirty four weeks. Late preterm steroids can be considered up to thirty six weeks and six days only in selected patients who match the late preterm logic: singleton pregnancy, risk of delivery before thirty seven weeks, no pregestational diabetes, and no prior steroid course.

But infection changes everything.

If chorioamnionitis is suspected, the pregnancy is no longer a steroid-completion project. Infection is a delivery problem. The fetus does not benefit from remaining in an infected environment simply so the clock can finish.

Now add tocolysis.

Tocolysis is often described casually as "stopping contractions." That phrase is dangerous. Tocolysis does not treat the underlying reason for preterm birth. It buys time.

What time?

Time for betamethasone. Time for magnesium neuroprotection when indicated. Time for GBS prophylaxis. Time for transfer to a center with the right neonatal resources.

That is a legitimate purpose. But a time-buying drug is only appropriate when time is worth buying.

In PPRM, tocolysis may be considered selectively to complete betamethasone if there are no signs of infection. But tocolysis is not a long-term strategy, and it should not be used to override infection, bleeding, fetal compromise, or severe maternal disease.

The resident should ask, what exactly are we buying, and what risk are we accepting to buy it?

Now consider the interaction between steroids, tocolysis, fluids, and maternal physiology.

Beta-mimetic tocolytics can worsen maternal hyperglycemia and hypokalemia. Steroids worsen glucose control. Multiple gestation, steroids, tocolysis, and intravenous fluids can increase pulmonary edema risk. Preeclampsia itself also increases pulmonary edema risk.

This is why a steroid plan is never just a fetal plan. It is also a maternal safety plan.

Now move to preeclampsia and H E L L P.

Steroids in preeclampsia are often misunderstood because they appear in two different conversations.

Conversation one is fetal maturation. If a patient with preeclampsia is likely to deliver preterm before thirty four weeks, and maternal and fetal status allow time, antenatal corticosteroids may help the fetus.

Conversation two is maternal treatment of H E L L P. High-dose corticosteroids may transiently improve platelet counts or laboratory values, but the major guideline logic in the corpus does not support steroids as definitive treatment for H E L L P because clear maternal outcome benefit is not proven. Maternal stabilization and delivery remain the treatment framework.

These conversations must not be mixed.

In preeclampsia, steroids may help the preterm fetus. They do not make severe maternal disease safe.

If severe features are present at thirty four weeks or beyond, delivery after maternal stabilization is the usual logic. Delivery should not be delayed simply to give late preterm steroids. If severe disease appears before thirty four weeks, expectant management is only for carefully selected stable patients in appropriate centers. Steroids may be part of that plan, but they are not the reason to ignore deterioration.

Now let us test the steroid-completion trap.

Case one. Thirty one weeks, PPRM, no fever, no uterine tenderness, reassuring fetal status, no labor requiring immediate delivery. Betamethasone is appropriate. A brief time-buying strategy may be reasonable if contractions threaten delivery before the course is complete and there is no infection.

Case two. Thirty one weeks, PPRM, maternal fever, uterine tenderness, fetal tachycardia. This is not a steroid-completion scenario. This is suspected infection. Delivery and antibiotics move to the front.

Case three. Thirty three weeks, severe preeclampsia with worsening platelets and pulmonary edema. A steroid dose may be considered if it does not delay care, but the mother is declaring danger. Steroid completion is not the anchor.

Case four. Thirty two weeks, preterm labor in a twin pregnancy, receiving fluids, with shortness of breath and rising oxygen need. Before adding more tocolysis to finish steroids, think pulmonary edema. The maternal cardiopulmonary state may be the limiting factor.

The resident reflex is this: complete the steroid course only if time is safe.

Ask four questions.

Is there infection?

Is the mother stable?

Is the fetus stable?

Is the thing we are doing to buy time safer than delivery now?

If those answers are reassuring, steroid completion can be an excellent goal. If they are not, steroid completion becomes a trap.

The steroid course is powerful because it improves neonatal transition. But it does not neutralize chorioamnionitis, severe preeclampsia, pulmonary edema, abruption, or fetal compromise.

Good obstetrics knows when to buy time and when time has become the danger.